

Rufunsa Town Council

Notebook and Kaggle Documentation

Data workflow, provenance, publication guidance, and data-quality notes

1. Project purpose

This project builds reusable datasets from Rufunsa Town Council publications. The work centres on Constituency Development Fund (CDF) community-project submissions and tables extracted from the Rufunsa Integrated Development Plan (IDP). It is intended to make source material easier to inspect, analyse, and publish as an open dataset.

The main workflow is implemented in notebooks/Untitled.ipynb. The notebook creates the project directories, records source links, downloads source files, extracts tables, cleans records, adds provenance fields, and writes pipe-delimited CSV files.

2. Repository layout

Location	Contents	Status
notebooks/Untitled.ipynb	96-cell Python notebook (80 code, 16 Markdown)	Primary workflow; contains exploratory and production cells
data/raw/rufunsa_sources.csv	Inventory of 8 official council sources	Pipe-delimited
data/raw/rufunsa_cdf_links.csv	9 filtered CDF links discovered from the council CDF page	Pipe-delimited
data/raw/2025_cdf_projects_received_from_wdcs.xlsx	Original 2025 WDC community-project submission workbook	Source file
data/raw/rufunsa_idp.pdf	Rufunsa Integrated Development Plan PDF	Source file
data/raw/2025_approved_secondary_school_bursaries.pdf	Downloaded bursary-related PDF	Source file; see known limitations
data/cleaned/db-unza26-csc4792-rufunsa_cdf_community_projects.csv	Curated 2025 community-project dataset	Analysis-ready, subject to data-quality notes
data/cleaned/db-unza26-csc4792-rufunsa_idp_projects.csv	Tables extracted from the IDP PDF	Broad raw extraction; not a project-only table

All CSV files use a pipe (|) delimiter. Read them with `pd.read_csv(path, sep="|")`.

3. Data sources and provenance

The notebook records the following official sources in `rufunsa_sources.csv`:

1. Rufunsa Town Council website.
2. Rufunsa Town Council CDF page.
3. WDC Community Project Submissions 2024.
4. Rufunsa Integrated Development Plan, 2024–2034.
5. Rufunsa Constituency and Wards page.

6. Council publications page.
7. Civic leaders page.
8. Council departments page.

The community-project dataset is derived from the council workbook “2025 CDF Projects Received from WDCs.” Each exported record retains the source URL, source document title, and retrieval date so that results can be traced back to the publication.

4. Notebook workflow

A. Set up folders and source inventory

The first cells create data/raw, data/cleaned, and documents. They then construct a small source catalogue and save it to data/raw/rufunsa_sources.csv.

B. Discover CDF publications

The notebook requests the Council CDF page, parses its links with BeautifulSoup, filters links matching CDF-project, bursary, empowerment, skills-development, road, and approval titles, and saves the result in data/raw/rufunsa_cdf_links.csv.

C. Build the 2025 CDF community-project dataset

The notebook selects the 2025 CDF Projects Received from WDCs link, downloads its Excel workbook, and identifies the actual header row (row 4 when loaded with header=None). It then:

- removes wholly empty rows and columns;
- removes duplicate records;
- normalises column names;
- renames key fields such as no. to project_id and number_of_beneficiaries to beneficiaries;
- adds council, province, district, constituency, financial year, source, and retrieval metadata; and
- exports the result to data/cleaned/db-unza26-csc4792-rufunsa_cdf_community_projects.csv.

D. Explore bursary, skills, and empowerment sources

The next section identifies relevant CDF links and demonstrates requests and PDF-table extraction with pdfplumber. It defines target fields for bursaries and empowerment, but it is not yet a completed, reliable export workflow.

E. Extract IDP tables

The IDP section downloads the PDF, extracts text and every detected table with pdfplumber, concatenates the table fragments, adds geographic metadata, validates basic shape/duplicates, and writes db-unza26-csc4792-rufunsa_idp_projects.csv.

5. Published CDF community-project dataset

db-unza26-csc4792-rufunsa_cdf_community_projects.csv contains 96 records and 20 columns.

Field	Meaning
project_id	Identifier/sequence from the source workbook
ward	Ward where the project is proposed
zone	Zone/locality reported in the source
project_name	Name of the proposed project
project_description	Narrative project description

Field	Meaning
sector	Reported sector
project_type	Reported project type
beneficiaries	Beneficiary count or source text; not consistently numeric
cdf_application_form_yes_no	Whether a CDF application form was reported
wdec_recommendation_yes_no	Whether WDEC recommendation was reported
wdc_minutes_yes_no	Whether WDC minutes were reported
zonal_minutes_yes_no	Whether zonal minutes were reported
council, province, district, constituency	Geographic metadata added by the notebook
financial_year	Financial year (2025)
source_url	Direct source-workbook URL
source_document	Source document title
retrieval_date	Date the notebook wrote the dataset

Snapshot of the CDF data

- The largest reported sector is Education (46 records), followed by Water (20) and Health (18).
- The dataset includes 13 distinct ward spellings/values. Chitimbwi and Chitimbwi differ only by trailing whitespace and should be standardised before ward-level totals are calculated.
- The beneficiaries column is mixed-format: most values are numeric, while some are blank, nill/Nill, or text such as 12 households. Treat it as text until a clearly documented numeric-cleaning rule is applied.
- Sector and project-type labels also have inconsistent whitespace and wording (for example, Education and Education). Normalise these values before aggregation.

The document values reflect the files present in this repository on 14 September 2026 and are not claims of current project approval, funding, or completion.

6. IDP extract: important interpretation note

The IDP output has 2,037 rows and 16 columns. It contains all tables detected in the PDF, including demographic and other plan tables, rather than a curated list of IDP projects. Several columns are positional names (0, 1, ...) because the PDF tables do not share a single consistent schema.

For Kaggle, publish this file as “Rufunsa IDP PDF Table Extract”, not as “IDP Projects,” unless it is further filtered and transformed into the target project schema:

record_id, council, province, district, constituency, ward, project_name, sector, project_description, implementation_period, project_status, estimated_cost, source_url, source_document, retrieval_date.

7. Kaggle publication guide

Recommended dataset title

Rufunsa 2025 CDF Community Project Submissions

Suggested Kaggle description

This dataset contains 96 community-project submissions received from Ward Development Committees for the 2025 Constituency Development Fund cycle in Rufunsa, Lusaka Province, Zambia. It was transformed from an official Rufunsa Town Council workbook. The dataset preserves source URL, document name, and retrieval date for provenance. It represents submissions received and should not be interpreted as a list of approved, funded, or completed projects.

Files to upload

1. db-unza26-csc4792-rufunsa_cdf_community_projects.csv — recommended primary dataset.
2. rufunsa_sources.csv — recommended supporting provenance file.
3. rufunsa_cdf_links.csv — optional link catalogue.
4. db-unza26-csc4792-rufunsa_idp_projects.csv — optional, only when labelled as a raw PDF table extract and accompanied by the limitation note above.

Do not upload the source PDFs/XLSX unless their reuse terms and personal-data implications have been reviewed.

Tags and usability notes

Suggested tags: zambia, public-sector, local-government, community-development, open-data, and infrastructure.

Set the dataset type to tabular data. State that values are sourced from a public local-government publication, that the data is not independently verified, and that the original source should be consulted for official decisions.

Kaggle notebook starter

```
import pandas as pd

cdf = pd.read_csv(
    "/kaggle/input/rufunsa-2025-cdf-community-project-submissions/"
    "db-unza26-csc4792-rufunsa_cdf_community_projects.csv",
    sep="|"
)

# Standardise category fields before analysis
for col in ["ward", "sector", "project_type"]:
    cdf[col] = cdf[col].astype("string").str.strip()

# Preserve the raw beneficiary field; derive a numeric field safely.
cdf["beneficiaries_numeric"] = pd.to_numeric(
    cdf["beneficiaries"], errors="coerce"
)

projects_by_sector = cdf["sector"].value_counts(dropna=False)
projects_by_ward = cdf["ward"].value_counts(dropna=False)
```

8. Requirements and reproduction

The notebook uses Python plus pandas, requests, beautifulsoup4, pdfplumber, and an Excel engine supported by pandas (normally openpyxl).

Run the notebook from the notebooks directory so paths such as ../data/raw resolve to the repository's data/raw folder. Internet access is needed for the download cells; the existing raw files allow later cleaning and analysis to be repeated without downloading them again.

For reproducibility, avoid using `verify=False` in production requests unless there is a documented certificate-validation issue. Capture package versions in a `requirements.txt` or Kaggle environment specification before publishing a fully reproducible notebook.

9. Known limitations and recommended next work

1. A generic early download block refers to `file_type` before it is set; it needs parameters or removal.
2. The bursary/skills/empowerment section mixes source variables and filenames (for example, it downloads from `skills_url` while saving a bursary filename) and later uses variables that are not defined in the notebook (`bursary_url`). It should be rebuilt as separate, explicit pipelines for each source.
3. The PDF-table code has an indentation issue in one conditional block and reuses `all_tables` across sections. Use scoped variables such as `bursary_tables` and `idp_tables`.
4. The IDP output needs a table-selection, header-detection, and field-mapping stage before it can support project-level analysis.
5. Add validation rules for IDs, ward names, categorical labels, beneficiary values, source URLs, and duplicate keys. Keep the original raw source unchanged and publish a separate data-cleaning log.

10. Citation

When using the primary dataset, cite:

Rufunsa Town Council. (2025). 2025 CDF Projects Received from WDCs [Excel workbook]. Retrieved through the Rufunsa Town Council CDF page. Processed in the CSC4792 Rufunsa project.

Also cite the `direct_source_url` included in the dataset and record the version/retrieval date used in the analysis.